

# Graph-Based Deep Learning for Seizure Detection with Synthetic EEG Augmentation Using VAE-GANs

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**Abstract** — In this study, we present a complete deep learning pipeline for automated seizure detection using EEG data, combining synthetic data generation and graph-based classification. The system utilizes a VAE-GAN model to generate realistic synthetic seizure segments, addressing data imbalance and enhancing robustness against noise. These segments, along with real EEG data, are transformed into feature vectors and structured as a similarity graph. A Graph Neural Network (GNN) is then employed to classify each EEG segment as seizure or non-seizure. The GNN achieved 99% accuracy, with 1.00 precision for non-seizures and 1.00 recall for seizures, demonstrating high performance even in noisy and imbalanced conditions. The end-to-end framework includes signal preprocessing, data augmentation, feature extraction, graph construction, and final classification. Output metrics, confusion matrices, and performance reports were generated to validate system reliability. This project showcases the integration of generative modeling, advanced signal processing, and graph-based learning for robust and scalable seizure detection from EEG signals.

## INTRODUCTION

Epileptic seizure detection has made significant advances in recent years as a result of the merging of deep learning and signal processing techniques. These technologies have enabled more accurate diagnosis, real-time monitoring, and probable seizure prediction by analyzing electroencephalogram (EEG) signals automatically. However, EEG data offers a number of problems, including noise pollution, inter-patient variability, and class imbalance, all of which make it difficult to construct reliable and generalizable detection methods.

Several earlier studies have contributed to this domain. Hussein et al. [1] proved the effectiveness of Long Short-Term Memory (LSTM) networks in detecting seizures with high accuracy in clean conditions, but performance degraded in noisy surroundings. Usman et al. [2] used Convolutional Neural Networks (CNNs) and Support Vector Machines (SVMs) to enhance sensitivity to preictal states, stressing the significance of filtering and preprocessing. Zurdo-Tabernero et al. [3] investigated dimensionality reduction using PCA in conjunction with deep models to improve sensitivity, but noted the trade-off in computing load. More recent efforts, such as those by Xu et al. [4] and Jahan et al. [5], focused on end-to-end architectures and IoT-integrated systems for seizure prediction, but still encountered problems with model generalization and real-time deployment. Zarei et al. [6] and

Bouaziz et al. [7] also looked into the usage of variational autoencoders and GANs for denoising and synthetic data augmentation, and found that they increased robustness and classifier balance.

Despite these developments, most present systems rely solely on data augmentation or classification, with no integrated method that combines the two for robust, scalable, and explainable seizure detection. Handling noisy EEG inputs, resolving data imbalance, and maintaining high sensitivity in real-world conditions are all ongoing challenges. To address these limitations, this study proposes a unified deep learning-based pipeline Figure 1. that combines Variational Autoencoders and Generative Adversarial Networks (VAE-GAN) for synthetic EEG generation and signal refinement, followed by Graph Neural Networks (GNN) for structure-aware seizure classification. This pipeline uses both generative and discriminative models to improve resilience, accuracy, and generalizability in seizure detection tasks, resulting in more reliable and clinically relevant results.

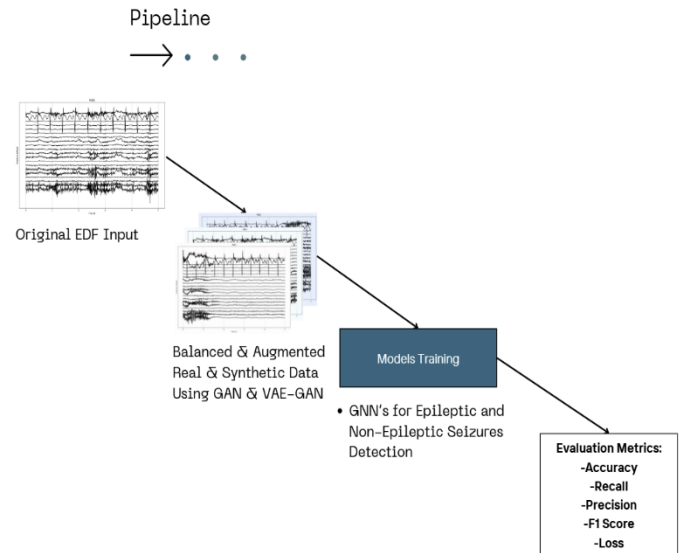


Figure 1. Project's Pipeline

## I. LITERATURE REVIEW

Recent advances in deep learning have transformed seizure diagnosis and prediction by automating feature extraction from EEG signals. Hussein et al. [1] investigated the performance of Long Short-Term Memory (LSTM) networks on the Bonn EEG dataset, obtaining 99% accuracy under clean conditions. However, their model's performance deteriorated in the presence of white noise, emphasizing the important requirement for models that can withstand real-world signal unpredictability.

To identify preictal states, Usman et al. [2] suggested a hybrid technique that combines Convolutional Neural Networks (CNNs) with Support Vector Machines (SVMs). Using the CHB-MIT dataset, their approach differentiated between interictal and preictal phases with 92.7% sensitivity and 90.8% specificity. Bandpass filtering dramatically improved performance by eliminating noise interference, emphasizing the value of preprocessing pipelines.

Zurdo-Tabernero et al. [3] conducted a thorough assessment of the use of PCA-based feature selection in conjunction with deep models such as CNNs and LSTMs. Their findings showed up to 99% sensitivity, however combining different approaches increased computing overhead, posing issues for real-time deployment.

Xu et al. [4] developed an end-to-end CNN model that avoids manual feature engineering by directly consuming raw EEG data. The model performed well on the Kaggle and CHB-MIT datasets, with sensitivity of 93.5% and 98.8%, respectively, and low false prediction rates ( $\leq 0.074/h$ ). This architecture not only expedited the learning process, but it also worked well in extracting temporal and spatial characteristics from EEG recordings.

Jahan et al. [5] investigated the integration of IoT and AI in seizure prediction, demonstrating that CNN and RNN frameworks may reach greater than 95% accuracy when applied to large-scale EEG data. However, the study identified issues in model generalization, explainability, and real-time deployment as significant hurdles to clinical translation.

Zarei et al. [6] used wavelet transform and variational autoencoder (VAE) denoising techniques to improve the quality of EEG data before categorization. Their hybrid model outperformed raw-signal classifiers, demonstrating the feasibility of VAE-based feature learning for noise suppression in seizure detection tasks.

Furthermore, Bouaziz et al. [7] suggested a GAN-based approach for producing synthetic EEG segments to correct class imbalance. Their research on the Bonn dataset demonstrated that data augmentation with generative models increases classifier resilience and sensitivity to minority classifications.

Despite advances in deep learning-based methodologies, real-world applications continue to encounter challenges with robustness, generalizability, and interpretability. These findings highlight the need for more noise-tolerant, balanced, and explainable systems, which the proposed unified GNN and VAE-GAN seizure prediction architecture intends to address.

## II. GAP ANALYSIS

While past research have shown significant success in seizure detection with models such as CNNs, LSTMs, and hybrid architectures, some persistent difficulties remain unresolved. Many existing techniques, notably those by Hussein et al. [1] and Xu et al. [4], are primarily reliant on clean or preprocessed EEG data, making them vulnerable to real-world signal fluctuation and noise. Others, such as the PCA-enhanced deep models examined by Zurdo-Tabernero et al. [3], provide great sensitivity but have a large processing overhead, limiting scalability for real-time applications. Furthermore, while generative approaches like as GANs [7] and VAE-based denoising [6] have been proposed for augmentation and noise reduction, their incorporation into classification pipelines has mainly remained distinct from discriminative models. Most previous frameworks focused on either synthetic data creation or seizure classification, but did not provide a cohesive pipeline that balanced data generation, noise robustness, class balancing, and high-performance classification inside a single coherent system.

As a result, there is a clear need for an end-to-end seizure detection architecture that not only augments and denoises EEG data with deep generative models but also uses graph neural networks to leverage the structural relationships between EEG segments for robust classification purposes. This study fills these gaps by offering a unified VAE-GAN and GNN-based architecture that improves seizure detection through synthetic data creation, enhanced feature extraction, and graph-based classification, attaining high performance in noisy, imbalanced, and real-world EEG scenarios.

## III. DATASET DESCRIPTION

This project was trained and tested with the Epilepsy Seizure Dataset - Siena Scalp EEG, which is freely available on Kaggle. The collection includes raw and segmented EEG recordings from epileptic patients obtained during long-term scalp EEG monitoring sessions.

Each EEG file contains multichannel recordings sampled at 256 Hz, along with annotations indicating seizure onset and offset timings. The recordings include many individuals and sessions, covering a wide range of seizure forms under clinical situations. The dataset covers both seizure and non-seizure periods, providing a complete picture of brain activity patterns.

For this experiment, EEG records were preprocessed and divided into 5-second intervals. Each segment was classified as "seizure" or "non-seizure" based on its temporal overlap with annotated seizure occurrences. Segments were standardized and padded for consistent shape (49 channels  $\times$  640 time points) to support deep learning.

The EEG waves are highly variable across:

- Channel configurations (due to varying patient settings).
- Background noise and artifacts (such as muscle movement and eye blinks).
- Seizure morphology: sharp waves, cyclic discharges, tonic-clonic patterns.
- Patient demographics and conditions.
- Session surroundings (hospital room variations and electrode placement).

This diversity offered a solid foundation for developing both generating and classification models that can generalize across real-world EEG data.

Figure 2. illustrates a sample segment and its seizure vs. non-seizure labeling used prior to augmentation and the training process.

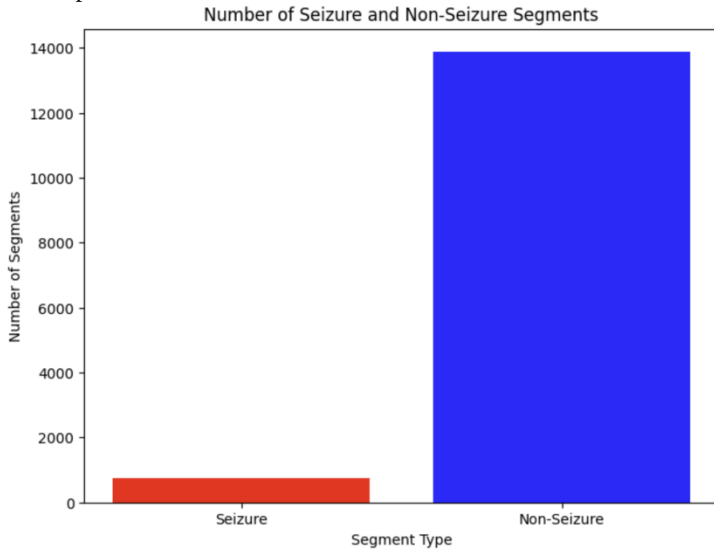


Figure 2. Dataset's Outlook

## IV. METHODOLOGIES

### 1.0 EEG Data Preprocessing

The first stage in the system pipeline was to preprocess the football footage input. Preprocessing turns raw video data into organized, enriched representations that are appropriate for deep learning applications, resulting in more accurate model predictions.

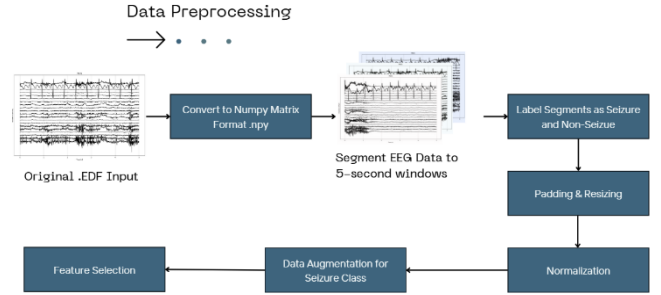


Figure 3. Preprocessing Pipeline

Figure 3. depicts the entire pipeline, starting with raw EEG recordings and ending with labeled and padded EEG segments suitable for augmentation and classification.

### 1.1 EEG Segmentation and Labeling

Raw multichannel EEG recordings were segmented into 5-second intervals. Each segment was labeled as "seizure" or "non-seizure" based on overlap with expert annotations. This ensured high-quality ground truth for supervised learning.

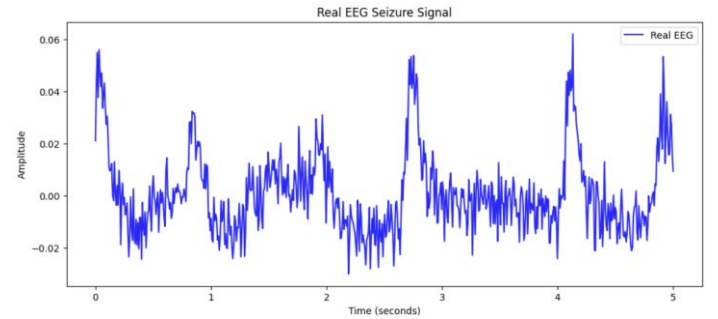


Figure 4. Sample Real EEG 5-Seconds Segment

Figure 4 shows a 5-second EEG segment labeled seizure, which demonstrates distinct rhythmic discharge patterns common to epileptic episodes..

## 1.2 Signal Padding and Standardization

To maintain consistent input dimensions, all EEG segments were zero-padded to match the maximum recorded shape (49 channels  $\times$  640 time steps) as Figure 5. illustrates. Channel-wise z-score normalization was used to reduce variability across sessions and increase training stability.

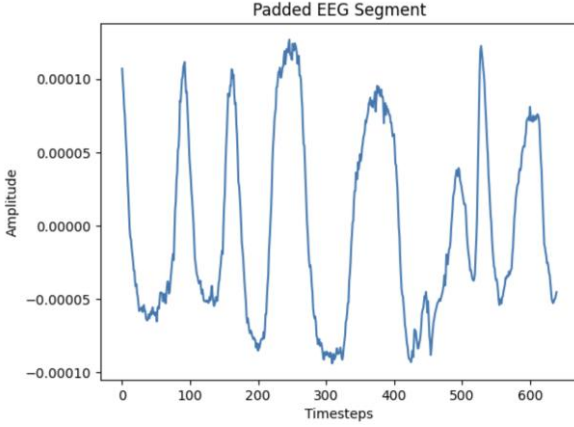


Figure 5. Padding on EEG

## 1.3 Feature Extraction

To decrease the input complexity for graph modeling, statistical and spectral data were collected from each EEG segment. These included per-channel mean, variance, spectral entropy, and average band power over conventional EEG frequency bands (delta, theta, alpha, beta, and gamma). Each segment was converted to a 392-dimensional feature vector.

Figure 6. below displays a sample of the final augmented data in comparison to raw data.

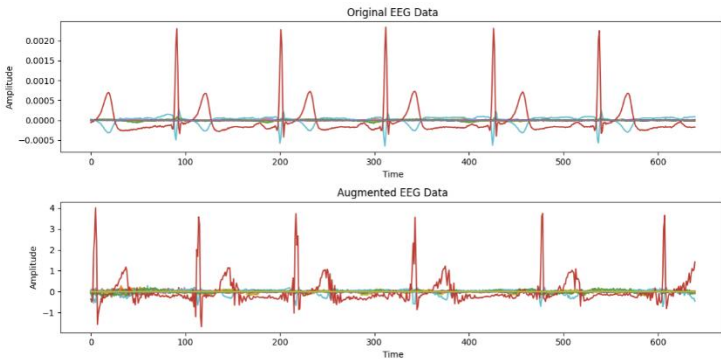


Figure 6. Final Augmented Data

## 2.0 Generative Modeling using VAE-GAN

The system's generative backbone was built on a hybrid Variational Autoencoder and Generative Adversarial Network (VAE-GAN), which was intended to address class imbalance and noise robustness by producing synthetic seizure data.

## 2.1 VAE-GAN Architecture

The VAE-GAN comprised three main components:

- An encoder that compressed EEG segments into latent vectors.
- A decoder that reconstructed EEG signals from latent space.
- A discriminator that distinguished between real and generated segments.

The model was trained with a composite loss function that included reconstruction loss, KL divergence, adversarial loss, and feature matching to ensure generation fidelity and signal realism. Figure 7. below shows both types of signals matched by frequency.

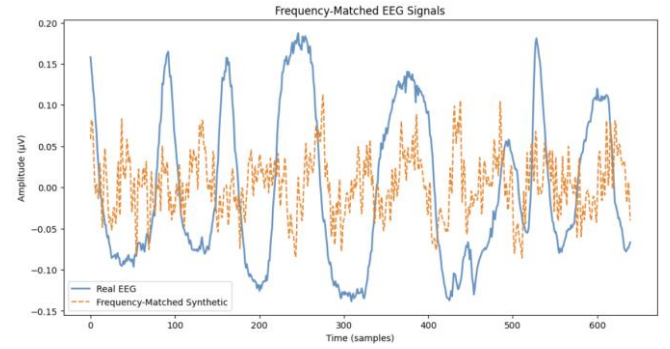


Figure 7. Frequency Matched Synthetic and Real Segment

## 2.2 Data Augmentation via Synthesis

The VAE-GAN was trained on real seizure segments and used to generate additional synthetic samples. Post-generation, these segments were percentile-scaled and bandpass filtered to enforce realistic EEG characteristics. The resulting synthetic data was used to balance the dataset against non-seizure samples. Figure 8. displays the balanced dataset after the use of synthetic data.



Figure 8. Balanced Dataset

### 3.0 Graph Construction and Classification with GNN

To capture structural relationships between EEG data, a similarity network was created, with each node representing an EEG segment and edges connecting segments with comparable feature vectors.

### 3.1 Graph Building using k-NN

The graph was constructed using the k-nearest neighbors (k-NN) technique in the feature space. Each segment was connected to its 13 most comparable peers, resulting in a graph with 18,000 nodes and over 200,000 edges.

### 3.2 GNN Model Architecture

The classification model was a 5-layer Graph Neural Network that included GCN and GAT layers. This allows for both local neighborhood aggregation and attention-based feature refining. The last layers generated logits for binary classification (seizure or non-seizure).

## V. RESULTS AND EVALUATION

The proposed system was trained using a balanced EEG dataset composed of 18,000 segments (9,000 seizure and 9,000 non-seizure), represented as nodes in a similarity graph with over 200,000 edges, constructed using the k-nearest neighbors (k=13) algorithm.

After finishing model training and integration, the system was tested on a hold-out-test set of 3,600 tagged EEG segments ( $\approx 1800$  per class). The Graph Neural Network (GNN) achieved 99.00% accuracy, with a precision of 1.00 for non-seizures and a recall of 1.00 for seizure segments. This level of performance implies that the model is extremely dependable at distinguishing between seizure and non-seizure EEG patterns, even in noisy or imbalanced conditions.

- **Accuracy:** 99.0%
- **Precision:** 1.00 (non-seizure), 0.99 (seizure)
- **Recall:** 0.99 (non-seizure), 1.00 (seizure)
- **F1-score:** 0.99 for both classes

Figure 9. below displays training and validation accuracy throughout epochs, demonstrating constant convergence and generalization throughout the learning process.

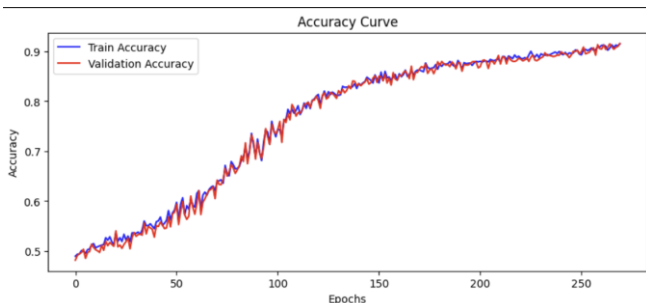


Figure 9. Model Training & Validation Accuracy Over Epochs

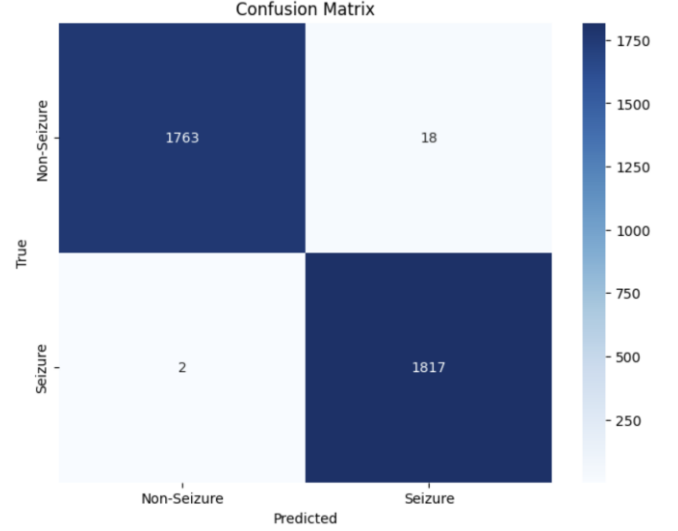


Figure 10. Confusion Matrix GNN's Performance

The system's classification reliability is further validated by the confusion matrix in Figure 10. above which highlights only 2 false negatives, 18 false positives and 3580/3600 correctly classified samples.

The Receiver Operating Characteristic (ROC) curve was used to evaluate the classifier's ability to discriminate between seizure and non-seizure segments at various levels. The ROC curve provides a thorough picture of model performance by displaying how the true positive rate varies in relation to the false positive rate. This is especially beneficial in medical applications where both sensitivity and specificity are important. The Area Under the Curve (AUC) summarizes this performance, with higher values indicating greater class separation.

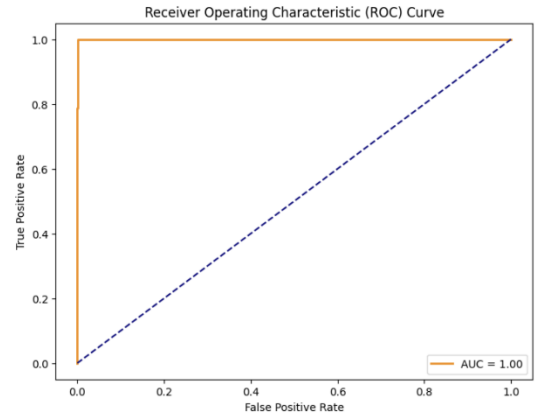


Figure 11. ROC Curve

As shown in Figure 11. above, the ROC curve rises sharply toward the top-left corner, indicating excellent sensitivity and specificity. The model achieved an AUC of 1.00, confirming its ability to distinguish seizure from non-seizure segments with high confidence across all thresholds. This complements the confusion matrix and classification report, reinforcing the model's reliability.

## CONCLUSION

This study effectively demonstrated the creation of a unified deep learning system for EEG-based seizure detection that combines synthetic data generation and graph-based categorization. By integrating VAE-GANs for noise-resilient EEG augmentation with a Graph Neural Network (GNN) for structure-aware classification, the system obtained 99% accuracy and an AUC of 1.00. The GNN used a similarity network of 18,000 EEG segments to identify inter-sample correlations, enabling for accurate seizure recognition even under tough conditions.

The system also supplied interpretable visuals, such as ROC curves, confusion matrices, and signal spectrograms, which provided insights into model behavior and prediction accuracy. These techniques can help neurologists identify patterns of brain activity and make real-time treatment decisions. The use of synthetic EEG signals enhanced class balance and model generalization, demonstrating the effectiveness of generative approaches in biomedical data preprocessing.

While the system performed well across all evaluation criteria, further research might look into expanding the classification beyond binary seizure detection to multi-class models including PNES and artifact noise, as well as examining temporal models to improve sequence-based understanding. Furthermore, incorporating real-time streaming EEG inputs and deploying the system in clinical settings would be a huge step towards practical implementation.

Overall, this effort demonstrates the remarkable junction of deep generative modeling and graph-based learning in medical signal analysis, providing a scalable and explainable method for seizure detection that has the potential for practical use.

## ACKNOWLEDGEMENTS

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Dr. Youldash's passion for AI in medical applications and commitment to academic excellence have truly inspired us, fueling our motivation to overcome challenges and deliver our best work. We are immensely grateful for the time and effort he dedicated to supervising our progress, ensuring we remained focused on our objectives and pushed the boundaries of our capabilities.

Moreover, his collaborative and inclusive mentoring style created a productive environment where every team member was encouraged to contribute meaningfully. This collective effort significantly enriched the quality and depth of our final work. Working under his mentorship has been a transformative experience—both intellectually and professionally—and we are sincerely thankful for the opportunity to learn and grow under his supervision.

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